

Direction, Form, and Strength in Scatter Plots

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

- | | |
|---|---|
| 1. the score at 5 hours = 74, — | 7. the median of the plotted values = 13.5, — |
| 2. the highest score plotted = 92, — | 8. the difference between the last and first values = 62, — |
| 3. the warmest hour = 14, — | 9. the mean price = 14000, — |
| 4. the mean pay = 72, — | 10. the value at the outlying point = 5, — |
| 5. the range of the plotted values = 9, — | |
| 6. the value at the unusual point = 30, — | |
-

What is verified

0 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.SP.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 20 tagged checks.

1. Revision hours against test score. Read the score at 5 hours, then describe the direction.

(a) Solution: Go to 5 on the horizontal axis and read up to the plotted point. It sits level with 74 on the score axis. 74

(b) Direction (instructor-judged). The association is positive: as revision time increases the scores increase too, so the cloud of points climbs from left to right. Accept any wording that names the direction and refers to both variables.

2. Television hours against test score. Give the highest score plotted, then describe the direction.

(a) Solution: Scan the vertical positions of the eight points; the top one is the student who watched no television. 92

(b) Direction (instructor-judged). The association is negative: the more hours of television, the lower the score, so the points fall from left to right. Accept any wording that names the direction and links it to both variables.

3. Temperature through the day. Give the warmest hour, then describe the form.

(a) **Solution:** The highest point on the plot is the one at hour 14, where the temperature reaches 28 degrees. 14

(b) **Form (instructor-judged).** The form is not linear. The temperature climbs to a peak in the early afternoon and then falls again, so the points make an arch. A straight line could follow the rising part or the falling part but not both. Accept any wording that rejects a straight line and describes the rise-then-fall shape.

4. Hours worked against pay. Find the mean pay, then describe the strength.

(a) **Solution:** Add the five amounts and divide by five:

$$\frac{24 + 48 + 72 + 96 + 120}{5} = \frac{360}{5}$$

72

(b) **Strength (instructor-judged).** The association is as strong as it can be: every point lies exactly on one straight line, with no scatter around it, so the hours worked predict the pay perfectly. Accept any wording that ties strength to how closely the points hug the line.

5. Shoe size against books read. Find the range, then describe the direction.

(a) **Solution:** The largest number of books is 14 and the smallest is 5, so the range is $14 - 5$. 9

(b) **Direction (instructor-judged).** There is no association. The points are scattered with no upward or downward trend, so knowing a student's shoe size tells you nothing about how many books they read. Accept any wording that says the direction is neither positive nor negative.

6. Practice hours against goals. Give the greatest number of goals, then circle the unusual point.

(a) **Solution:** The topmost point is the one at 7 hours of practice. 30

(b) **Outlier (instructor-judged).** The point to circle is the one at 7 hours with 30 goals. The other six points climb steadily by 2 goals per hour of practice; this one leaps far above that pattern, so it is an outlier. Accept any wording that names the point and compares it with the trend of the rest. Circling with no reason is only half the answer.

7. Adverts against orders. Find the median number of orders, then describe direction and strength.

(a) **Solution:** In order the eight values are 5, 8, 9, 13, 14, 18, 19, 24. With an even count the median is the mean of the middle two: $(13 + 14) \div 2$. 13.5

(b) **Direction and strength (instructor-judged).** The association is positive but only moderately strong. The points climb from left to right, but they zig-zag around the trend rather than sitting on a single line. A response must cover both features to earn full credit.

8. Day against bacteria colonies. Find the difference between day 7 and day 1, then describe the form.

(a) Solution: Day 7 shows 64 colonies and day 1 shows 2, so the difference is $64 - 2$.

62

(b) Form (instructor-judged). The form is not linear. The first four days rise gently by 2 each day and then the counts double, so the pattern bends sharply upward. Accept any wording that contrasts the gentle start with the steep finish.

9. Car age against price. Find the mean price, then describe direction, form and strength.

(a) Solution: Add the six prices and divide by six:

$$\frac{18000 + 16200 + 14600 + 13000 + 11800 + 10400}{6} = \frac{84000}{6}$$

14000

(b) Full description (instructor-judged). A complete answer names all three features: the direction is negative (older cars cost less), the form is linear (the points fall in a nearly straight line) and the strength is strong (the points sit very close to that line). Naming only one or two features earns partial credit.

10. Weeks against pages read. Give the value at the odd point, then describe the rest and its effect.

(a) Solution: Every other week sits between 20 and 42 pages and rises steadily; week 6 drops to the bottom of the plot.

5

(b) Description (instructor-judged). A correct response sets that point aside and describes the other six: they show a strong positive linear association, with the pages read climbing steadily from week to week. It then notes that the single low week makes the overall pattern look weaker and less regular than it really is, and that one unusual point can hide an otherwise clear trend. Accept any wording that separates the outlier from the pattern and describes both.